



Direct Memory Access

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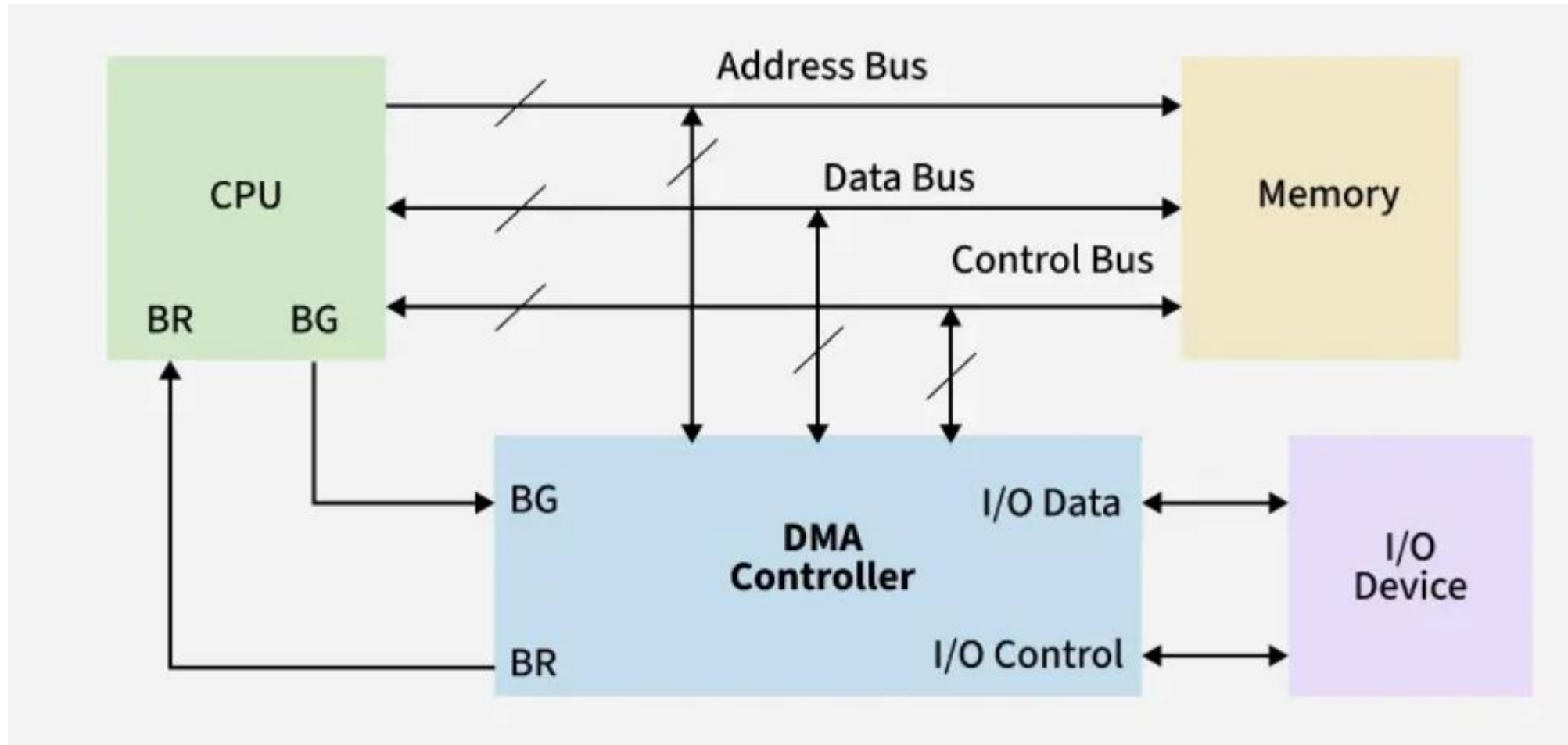
Direct Memory Access (DMA)

DMA allows hardware devices to transfer data directly to and from the computer's main memory without the CPU's involvement.

This significantly improving system performance, frees CPU.

A specialized hardware component called a DMA controller manages these transfers, freeing the CPU to perform other tasks.

This method is essential for high-speed data handling in peripherals like hard drives, graphics cards, and network cards.



How DMA works

1. Initiation:

A peripheral device or software needs to transfer data.

The CPU or device driver configures the DMA controller with the source and destination addresses and the amount of data to be transferred.

2. Bus control:

DMA controller requests control of the system's memory bus from the CPU.

After the CPU completes its current operation and acknowledges the request, the DMA controller takes over the bus.

3. Data transfer:

The DMA controller then manages the transfer of data directly between the peripheral device and the main memory. **This happens independently of the CPU.**

4. Completion

Once the data transfer is complete, the DMA controller signals it to the CPU

5. CPU resumes